INDEPENDENT COMPUTATIONAL STUDY

Reduced-Order Modelling of Resin Infusion in GFRP Composites

Literature Review, System Geometry, Governing Physical Assumptions, and Research Question Formulation

1. LITERATURE SYNTHESIS (REPORT INTRODUCTION DRAFT)

READY-TO-USE MANUSCRIPT INTRODUCTION PARAGRAPH

Liquid Composite Moulding (LCM) processes, such as Resin Transfer Moulding (RTM) and Vacuum-Assisted Resin Transfer Moulding (VARTM), fabricate structural glass fibre-reinforced polymer (GFRP) components by infiltrating a dry fibrous preform with a liquid thermoset resin. At the macroscopic scale, impregnation of a saturated porous preform is governed by Darcy's law, where the superficial resin velocity scales linearly with the pressure gradient and preform permeability K, and inversely with resin viscosity μ (Correia et al., 2005). Because preform permeability is an intrinsic microstructural property dictated by fibre architecture and packing density, Gebart (1992) derived closed-form micromechanical expressions linking axial and transverse permeability directly to filament radius R_f and fibre volume fraction V_f. While high fibre volume fractions ($V_f \geq 0.55$) are desirable to maximize composite stiffness and strength, they severely constrict interstitial pore channels, causing permeability to drop by orders of magnitude and risking incomplete mould filling prior to resin gelation. Furthermore, as highlighted by Michaud (2016) and Ruiz et al. (2006), variations in the advancing flow-front velocity alter the balance between viscous inter-tow flow and capillary intra-tow wetting. Consequently, reduced-order computational models that couple microstructural packing (V_f) with macroscopic Darcy flow kinetics provide a rapid, physically grounded tool to identify viable processing windows before conducting costly physical infusions.

2. CURATED LITERATURE MATRIX (KEY REFERENCES TO CITE)

Reference & Source	Core Scientific Contribution	Role in Your Technical Report
Gebart, B. R. (1992) *Permeability of Unidirectional Reinforcements for RTM*. J. Compos. Mater., 26(8), 1100–1133.	Derived analytical micromechanical models relating filament radius (R_f), periodic packing (quadratic/hexagonal), and fibre volume fraction (V_f) to axial ($K_{\parallel}$) and transverse ($K_{\perp}$) permeability.	Primary mathematical basis for coupling microstructural packing (V_f) to macroscopic Darcy permeability (K).
Michaud, V. (2016) *A Review of Non-saturated Resin Flow in Liquid Composite Moulding Processes*. Transp. Porous Med., 115(3), 581–601.	Comprehensively reviews deviations from classical saturated 1D Darcy flow in LCM, including capillary meniscus effects at the flow front, dual-scale fibrous porosity, and hydro-mechanical preform deformation.	Cite in *Introduction* and *Limitations* to demonstrate direct familiarity with EPFL LPAC's research framework on non-saturated vs. saturated flow.
Correia, N. C., et al. (2005) *Analysis of the Vacuum Infusion Moulding Process: I. Analytical Formulation*. Compos. Part A, 36(12), 1645–1656.	Establishes the analytical 1D Darcy formulation for rectilinear resin infusion under constant pressure differential and compares rigid-mould RTM against flexible-bag Vacuum Infusion (VI).	Justifies the 1D rectilinear domain geometry, boundary conditions (P_{in}, P_{vent}), and closed-form $x_f(t) \propto \sqrt{t}$ baseline solution.
Ruiz, E., et al. (2006) *Optimization of Injection Flow Rate to Minimize Micro/Macro-Voids Formation in RTM Composites*. Compos. Sci. Technol., 66(3-4), 475–486.	Demonstrates how local interstitial velocity $v_f = dx_f/dt$ and the modified capillary number (Ca^*) govern competition between viscous macro-flow (inter-tow) and capillary micro-flow (intra-tow).	Explains in *Discussion* why tracking dx_f/dt matters: overly fast flow fronts trap micro-voids inside fibre tows, while slow flow traps macro-voids.

3. DEFINED SYSTEM GEOMETRY & BOUNDARY CONDITIONS

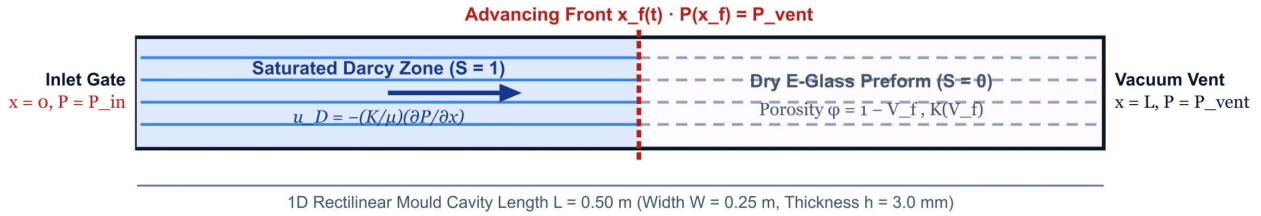


Figure 1: Schematic of 1D rectilinear resin infusion through a rigid E-glass fibrous preform under constant pressure differential $\Delta P = P_{in} - P_{vent}$.

- **Macroscopic Cavity Domain:** Rectilinear flat-plaque benchmark cavity (Arbter et al., 2011) of length $L = 0.50$ m along the flow axis ($x \in [0, L]$), width $W = 0.25$ m, and uniform thickness $h = 3.0$ mm. Full-width line-gate injection eliminates transverse pressure gradients ($\partial P/\partial y = \partial P/\partial z = 0$), reducing flow strictly to 1D along x.
- **Microstructural Architecture:** Unidirectional E-glass filaments of radius $R_f = 8.5\ \mu m$ ($17\ \mu m$ diameter) arranged in an idealized hexagonal periodic array with fibre volume fraction $V_f \in [0.35, 0.65]$ and interconnected open porosity $\varphi = 1 - V_f$.
- **Boundary & Initial Conditions:** (1) *Inlet ($x = 0$):* Constant injection pressure $P(0, t) = P_{in}$ for $t > 0$; (2) *Flow Front ($x = x_f(t)$):* Constant vent pressure $P(x_f, t) = P_{vent}$, establishing $\Delta P = P_{in} - P_{vent}$ across $0 \le x \le x_f(t)$; (3) *Initial Condition ($t = 0$):* Dry preform $x_f(0) = x_0 \to 0^+$ (initialized at $x_0 = 10^{-4}$ m in numerical ODE solvers to avoid the $t = 0$ singularity).

4. GOVERNING ENGINEERING ASSUMPTIONS

1. **Macroscopic Saturated Plug Flow (Sharp Flow-Front Approximation):** The partially saturated transition zone at the advancing front is neglected; the preform is fully saturated ($S = 1$) behind $x_f(t)$ and completely dry ($S = 0$) ahead of $x_f(t)$ (Michaud, 2016).
2. **Creeping Laminar Regime ($Re_p \ll 1$):** Pore-scale Reynolds numbers $Re_p = (\rho\, u_D\, R_f) / \mu$ remain well below 10^{-2}, ensuring viscous forces dominate over inertia and linear Darcy's law holds without Forchheimer quadratic drag corrections.
3. **Incompressible, Newtonian, Isothermal Fluid:** Liquid epoxy density ρ and dynamic viscosity μ remain constant in space and time ($t_{fill} \ll t_{gel}$), decoupling hydrodynamics from exothermic Castro-Macosko cure kinetics.
4. **Rigid, Stationary Fibre Preform:** Hydro-mechanical compaction and thickness relaxation typical of flexible vacuum bagging are neglected (V_f and $\varphi = 1 - V_f$ remain spatially and temporally uniform, representative of rigid-tool RTM or stiffened VARTM).
5. **Negligible Capillary Pressure Drop ($P_c \ll \Delta P$):** Under applied driving pressures $\Delta P \ge 0.5$ bar (50 kPa), macroscopic hydraulic pressure gradients dominate over capillary meniscus wicking pressure P_c.
6. **Single-Scale Homogenized Porosity:** Inter-tow and intra-tow pore spaces are homogenized into an effective macroscopic permeability tensor $K(V_f)$ predicted via Gebart's (1992) micromechanical formulation.

5. FORMAL RESEARCH QUESTION & SPECIFIC OBJECTIVES

PRIMARY RESEARCH QUESTION

"How does the microstructural fibre volume fraction (V_f) couple with macroscopic processing parameters—specifically resin viscosity (μ) and injection pressure (ΔP)—to govern flow-front progression kinetics ($x_f(t)$), interstitial velocity (dx_f/dt), and critical mould-filling time windows (t_{fill}) during 1D liquid composite moulding of E-glass/epoxy composites?"

1. **Model Formulation & Numerical Verification:** Formulate the 1D Darcy flow boundary-value problem coupled with Gebart's (1992) axial and transverse permeability equations, and verify the closed-form analytical